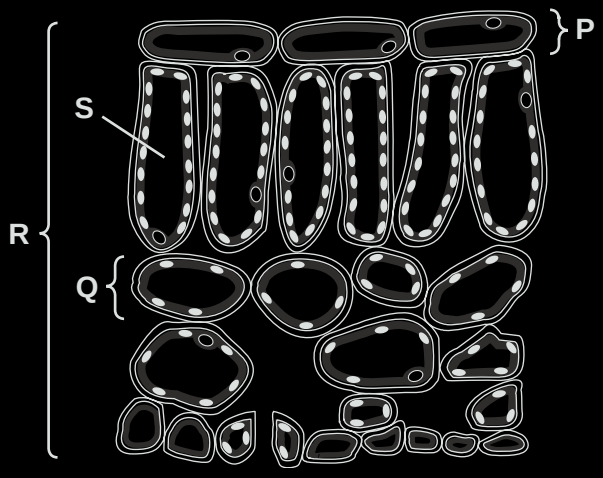


Which is an organ and which is a tissue?

	organ	tissue
A	**P**	**R**
B	**Q**	**S**
C	**R**	**P**
D	**S**	**Q**

2 What causes water to enter plant roots from the soil?

A Water potential in root hairs and the soil is equal.

B Water potential in root hairs and xylem is equal.

C Water potential in root hairs is higher than in the soil.

D Water potential in root hairs is lower than in the soil.

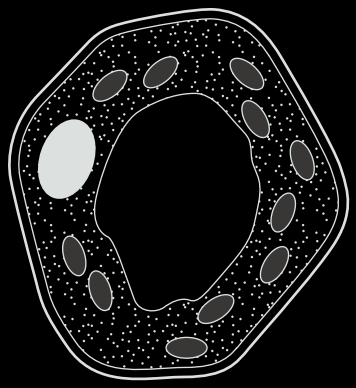
3 Which process may require energy from respiration?

A movement of carbon dioxide into the alveoli

B movement of oxygen into red blood cells

C uptake of glucose by cells in the villi

D uptake of water by root hair cells



What type of cell is it?

A epidermal cell of a leaf

B mesophyll cell of a leaf

C root hair cell

D xylem cell

8 Which chemical test shows the presence of an enzyme in a biological washing powder?

A Benedict's

B biuret

C ethanol emulsion

D iodine solution

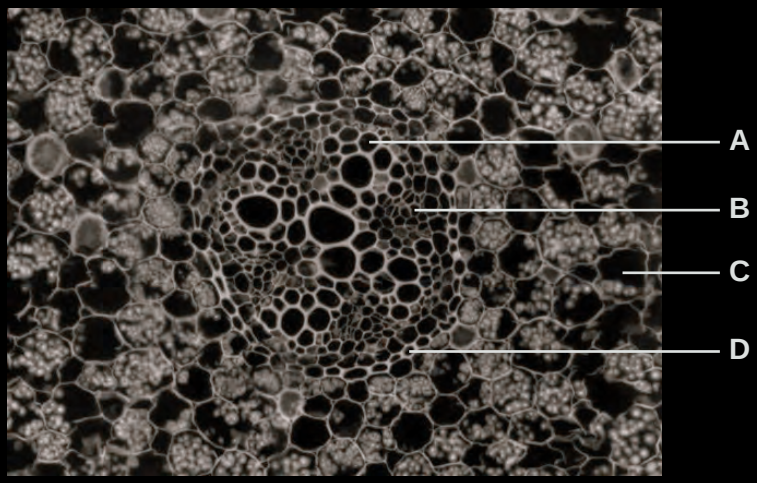
9 A lack of which nutrient causes gums to bleed?

A calcium

B iron

C vitamin C

D vitamin D



12 A young plant may wilt when dug up and re-planted in another place.

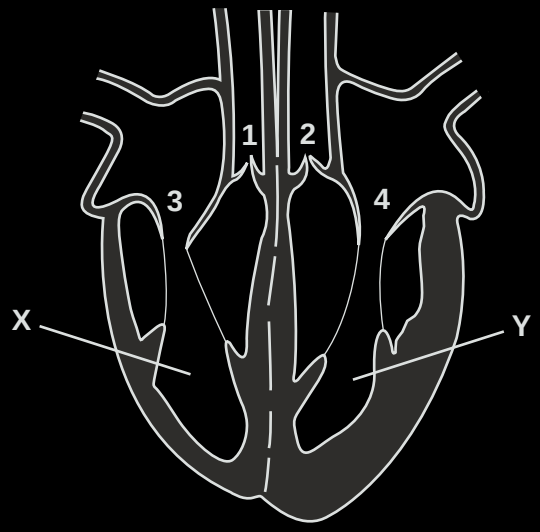
What causes this?

A The leaves lose less water.

B The roots cannot take up mineral salts.

C The stem cannot transport water.

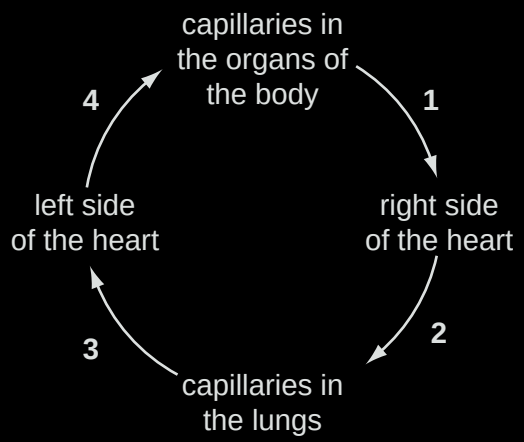
D The surface area of the root is reduced.



While chambers **X** and **Y** are emptying of blood, which valves are open and which are closed?

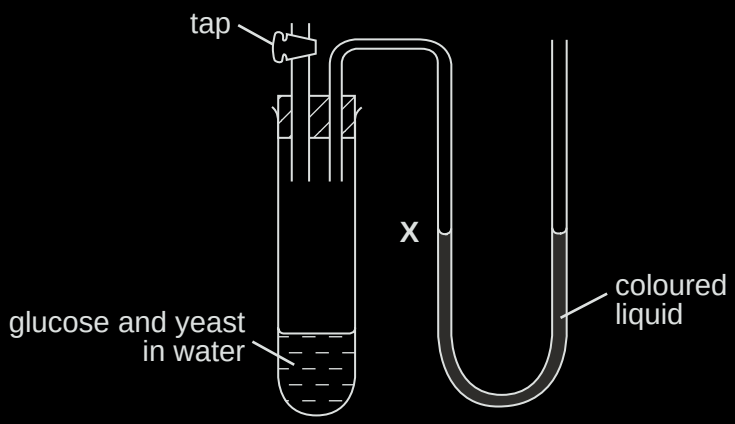
	valves **1** and **2**	valves **3** and **4**
A	closed	closed
B	closed	open
C	open	closed
D	open	open

14 The diagram represents blood flow through the human body.



At which stages does the blood contain the most oxygen?

A **1** and **2** **B** **2** and **3** **C** **3** and **4** **D** **4** and **1**



The tap is closed and the yeast respires anaerobically.

What is observed at **X** and which explanation is correct?

	observation at **X**	explanation
A	liquid level falls	carbon dioxide is produced
B	liquid level falls	oxygen is used
C	liquid level rises	carbon dioxide is produced
D	liquid level rises	oxygen is used

18 Fitness training increases the concentration of lactic acid that runners can build up in their muscles before pain stops them running.

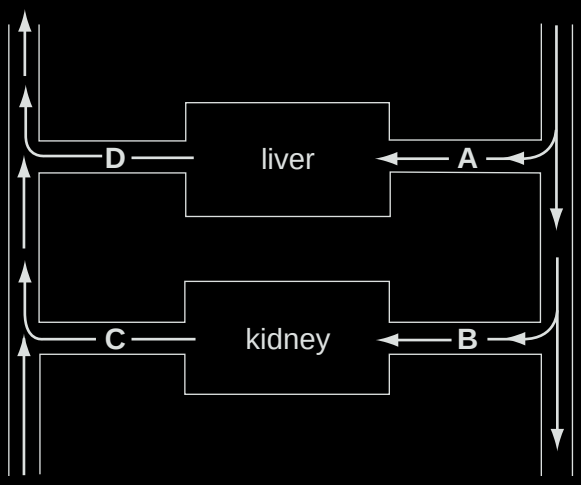
What is a consequence of this increase?

A Aerobic respiration in muscles can be more rapid.

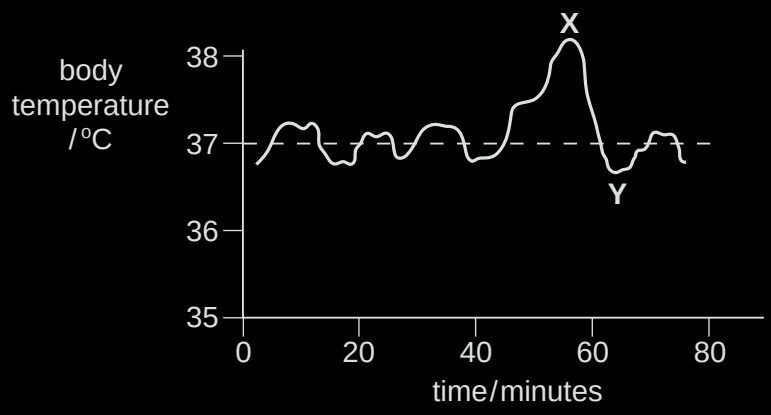
B Blood flow to the muscles is increased.

C More anaerobic respiration can take place in muscles.

D More energy is needed by the muscles.



20 The graph shows changes in a person's body temperature plotted against time.



What causes the change in temperature between **X** and **Y**?

A increased air temperature

B increased evaporation of sweat

C reduced blood flow through surface capillaries

D shivering

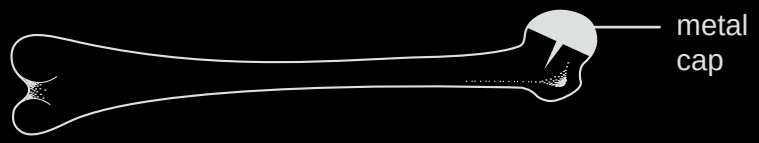
21 Why does emphysema cause severe breathlessness?

A The alveoli become coated with tar.

B The cilia lining the trachea are destroyed.

C The lungs become cancerous.

D The surface area of the lungs is reduced.



Which bone is this, and which joint does the metal cap repair?

	bone	joint
A	humerus	elbow
B	humerus	shoulder
C	ulna	elbow
D	ulna	shoulder

23 What is controlled by the medulla in the brain?

A balance

B breathing

C memory

D vision

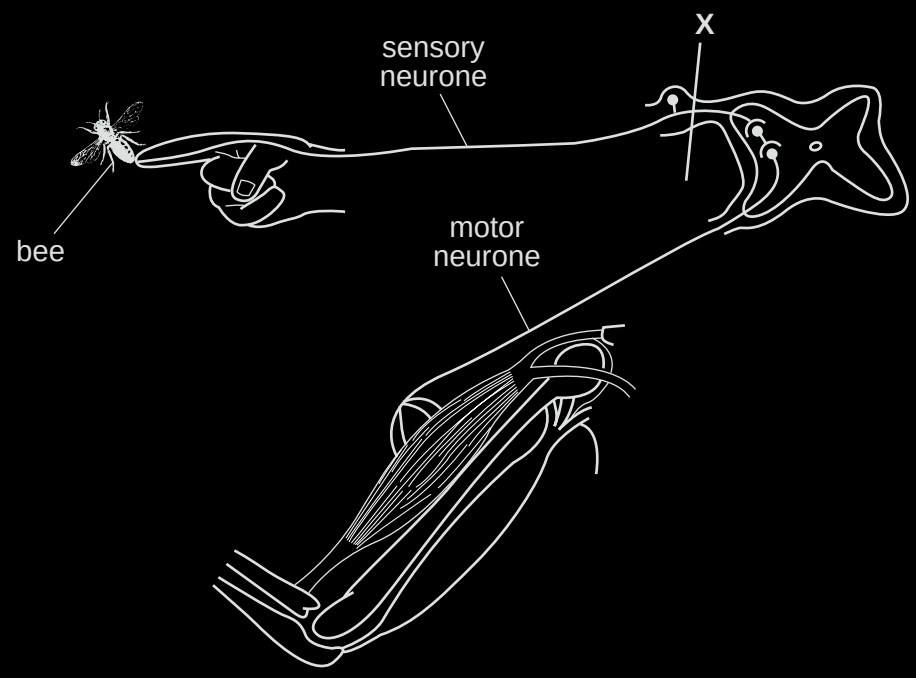
24 When a person is frightened, which substance increases the blood sugar levels?

A adrenaline

B carbon dioxide

C insulin

D lactic acid



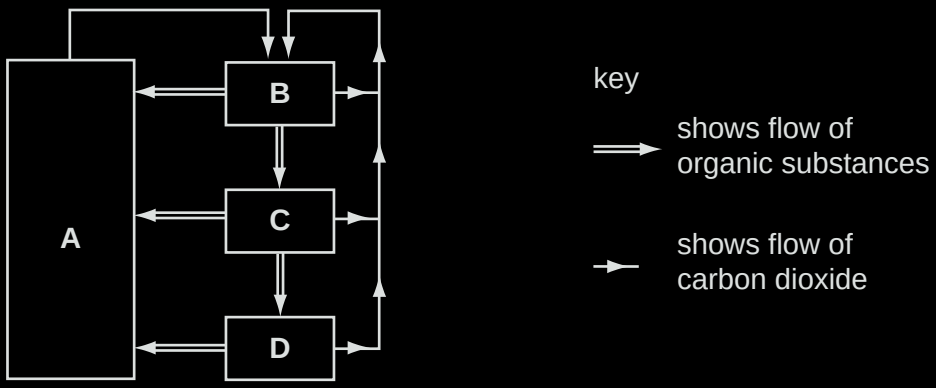
A bee stings the finger, as shown.

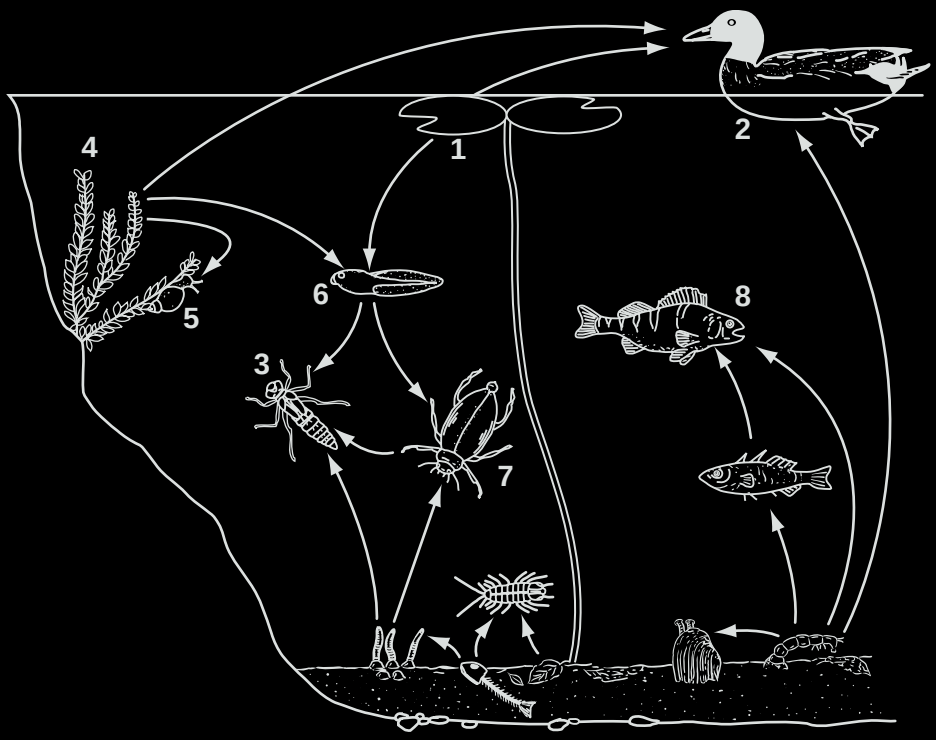
What are the effects of this sting on the person?

	pain felt	arm moved
A	no	no
B	no	yes
C	yes	no
D	yes	yes

26 When yeast is used in bread-making, what type of respiration occurs and which product is useful?

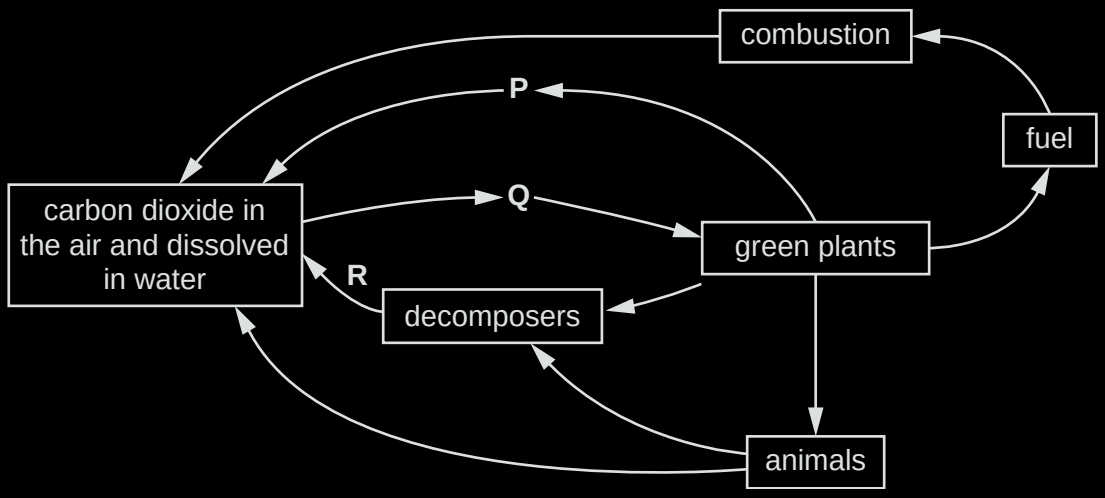
	respiration	useful product
A	aerobic	carbon dioxide
B	aerobic	ethanol
C	anaerobic	carbon dioxide
D	anaerobic	ethanol


A
B
C
D
key
shows flow of organic substances
shows flow of carbon dioxide



Which of the organisms is a producer, a herbivore or a carnivore?

	producer	herbivore	carnivore
A	**1**	**6**	**7**
B	**2**	**4**	**5**
C	**4**	**2**	**6**
D	**7**	**3**	**8**



What are processes **P**, **Q**, and **R**?

	P	Q	R
A	photosynthesis	photosynthesis	respiration
B	respiration	respiration	photosynthesis
C	photosynthesis	respiration	photosynthesis
D	respiration	photosynthesis	respiration

31 Pest control in hot countries includes these processes.

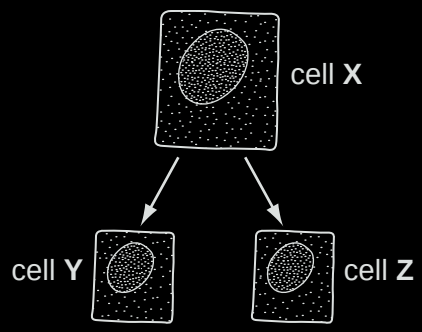
1 draining ponds
2 keeping cattle away from ponds
3 spraying oil on ponds

Which are used to control the malarial mosquito?

A **2** only **B** **1** and **2** **C** **1** and **3** **D** **1**, **2** and **3**

32 What will be the effect of increasing nitrate levels in rivers?

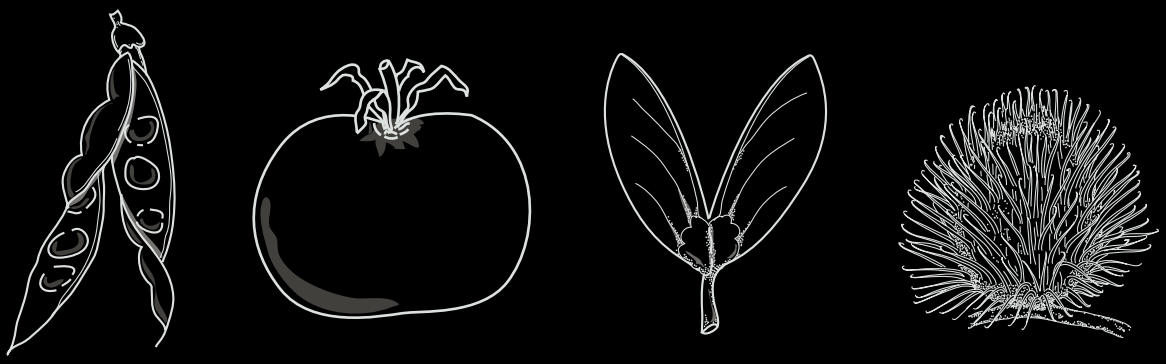
A Animals will absorb the nitrates and make more urea.
B Animals will absorb the nitrates and form more proteins.
C Plants will absorb the nitrates and make more urea.
D Plants will absorb the nitrates and form more proteins.



How many chromosomes does cell **Z** contain?

A 8 **B** 12 **C** 24 **D** 48

34 The diagrams show four different fruits.



1 **2** **3** **4**

Which two fruits are dispersed by animals?

A **1** and **2** **B** **1** and **3** **C** **2** and **4** **D** **3** and **4**

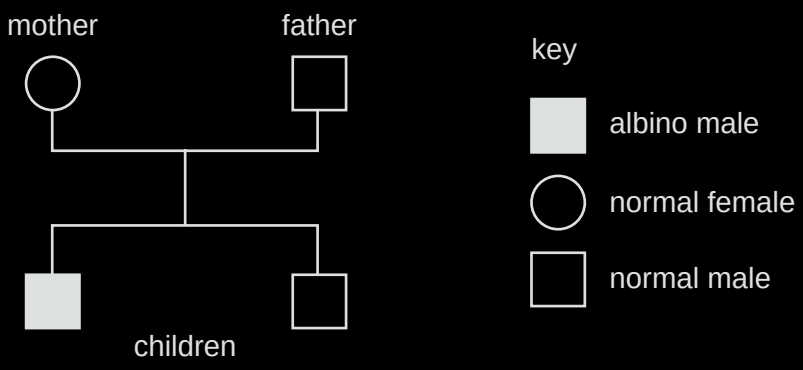
35 After sexual intercourse, sperm can survive for 3 days in the uterus and oviducts. Assume ovulation can occur any time from day 13 to day 15.

If an egg cell lives up to 2 days after ovulation, how long is the fertile phase of the menstrual cycle?

A 5 days **B** 6 days **C** 7 days **D** 8 days

36 What are the features of human eggs, when compared with sperm?

	size	number produced
A	larger	larger
B	larger	smaller
C	smaller	larger
D	smaller	smaller



What are the genotypes of the parents?

	mother	father
A	heterozygous	heterozygous
B	heterozygous	homozygous dominant
C	homozygous recessive	homozygous dominant
D	homozygous recessive	homozygous recessive

39 A person with Down's syndrome is born with 47 chromosomes in each of his/her cells, instead of 46.

What could cause this?

A A mutation happened during the production of the egg cell.

B More than one sperm fused with the egg at fertilisation.

C Radiation caused a change in structure of a gene in the father's sperm.

D The mother was exposed to harmful chemicals while she was pregnant.